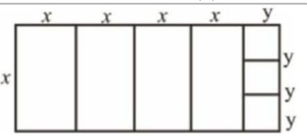
131. A big rectangular plot of area 4320 m^2 is divided into 3 square-shaped smaller plots y fencing parallel to the smaller side of the plot. However some area of land was still left as a square could not be formed. So, 3 more square-shaped plots were formed by fencing parallel to the longer side of the original plot such that no area of the plot was left surplus. What are the dimensions of the original plot?

(a) 160 m × 27 m (b) 240 m × 18 m
(c) 120 m × 36 m (d) 135 m × 32 m

Ans. (c)



Let the side of each large square be x and that of each small square be y

$\therefore 3(x^2) + 3(y^2) = 4320$

$x^2 + y^2 = 1440$ .....(i)

and area of original plot be

$x \times (3x + y) = 4320$ .....(ii)

be equ. (i) and (ii) we get

$3x^2 + xy = 3(x^2 + y^2)$

$xy = 3y^2$

$x = 3y$ .....(iii)

Now from equ. (i) and (iii)

$(3y)^2 + y^2 = 1440 \Rightarrow 10y^2 = 1440$

$y^2 = 144 \Rightarrow y = 12$

So $x = 36$

$\therefore$ Length of the plot = 3 × 36 + 12 = 120m

Width of the plot = x = 36 m.

132. 2 men and 1 woman board a bus in which 5 seats are vacant. One of these five seats is reserved for ladies. A woman may or may not sit on the seat reserved for ladies but a man cannot sit on the seat reserved for ladies. In how many different ways can the five seats be occupied by these three passengers?

(a) 15 (b) 36
(c) 48 (d) 60

Ans. (b) : Case 1 –

If lady sits on reserved seat then

2 men can occupy seats from 4 vacant seat in $^4P_2$ ways

= 4 × 3 = 12 ways

Case 2 –

If lady does not sits on reserved seat then the lady can occupy a seat from 4 vacant seat in 4 ways 1 and 1 man can occupy a seat from 3 seats in 3 ways and another man can occupy a seat from remaining 2 seats in two ways.

Total ways 4 × 3 × 2 = 24 ways

Hence from both the cases total number of ways = 12 + 24 = 36 ways

133. Which of the following pairs are correctly matched?

Publication		Group
1. Sportstar	:	The Hindu publication group
2. Business world	:	ABP group
3. The Week	:	Malayala Manorama publication group
4. Reader's Digest	:	Indian Express publication group

Select the correct answer using the codes given below:

(a) 1, 2, 3 and 4 (b) 1, 2 and 3
(c) 2 and 4 (d) 1 and 3

Ans. (b) : Readers Digest Magazine in India belongs to Times of India group. Rest is correctly matched.

134. Which of the following pairs are correctly matched?

Enterprise		Industrial Group
1. VSNL	:	Bharati Group
2. Mundra Special Economic Zone Ltd.	:	Adani group
3. CMC Ltd.	:	Tata Group
4. IPCL	:	Reliance Group

Select the correct answer using the codes given below:

(a) 1, 2 and 3 (b) 3 and 4
(c) 1, 2 and 4 (d) 2, 3 and 4

Ans. (d) : Videsh Sanchar Nigam Limited belongs to Tata group. Rest is correctly matched.

135. Match items in the List I (Business woman) with List II (Company) and select the correct answer using the codes given below the lists:

List I (Businesswoman)		List II (Company)
A. Zia Mody	1.	Venkateshware Hatcheries
B. Anuradha J. Desai	2.	AZB & Partners
C. Villoo Morawala Patell	3.	Quantum Market Research
D. Meena Kaushik	4.	Avestha Gengraine Technologies
	5.	Biocon India

Codes:

	A	B	C	D
(a)	4	1	5	3
(b)	2	3	4	1
(c)	4	3	5	1
(d)	2	1	4	3

Ans. (d) :

Zia Mody	AZB & Partners
Anuradha J. Desai	Venkateshware Hatcheries
Villoo Morawala Patel	Avestha Gengraine Technologies
Meena Kaushik	Quantum market research

136. A square is divided into 9 identical smaller squares. Six identical balls are to be placed in these smaller squares such that each of the three rows gets at least one ball (one ball in one square only). In how many different ways can this be done?

(a) 27 (b) 36
(c) 54 (d) 81

Ans. (d) :

137. There are 6 persons– A, B, C, D, E and F. They are to be seated in a row such that B never sits anywhere ahead of A and C never sits anywhere ahead of B. In how many different ways can this be done?

(a) 60 (b) 72
(c) 120 (d) None of the above

Ans. (d) :

138. For which one of the following items, is Tirupur well-known as a huge exporter to many parts of the world?

(a) Gems and Jewellery (b) Leather goods
(c) Knitted garments (d) Handicrafts

Ans. (c) : Tirupur in Tamil Nadu has gained universal recognition as the leading source of Hosiery, Knitted Garments, Casual Wear and Sportswear.

139. Which one of the following pairs is not correctly matched?

Project		Company
(a) Integrated Steel Plant at Jajpur (Orissa)	:	Steel Authority of India
(b) Power Plant at Jamnagar (Gujarat)	:	Essar Power
(c) Nabinagar Power Plant (Bihar)	:	Indian Railways
(d) Kayamkulam Power Plant (Kerala)	:	National Thermal Power Corporation

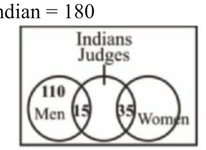
Ans. (a) : Integrated steel power plant at Jajpur, Orissa belongs to Tata group. Rest is correctly matched.

140. 300 persons are participating in a meeting, out of which 120 are foreigners and the rest are Indians. Out of the Indians, there are 110 men who are not judges; 160 are men or judges and 35 are women judges. There are no foreign judges. How many Indian women attended the meeting?

(a) 35 (b) 45
(c) 55 (d) 60

Ans. (c) : Foreigner = 120

Indian = 180



160 men or judge and 35 women judges

$\therefore$ men = 110 + 15 = 125

women = 160 – 125 = 55

141. There are 6 persons : A, B, C, D, E and F, A has 3 items more than C
D has 4 items less than B
E has 6 items less than F
C has 2 items more than E
F has 3 items more than D
Which one of the following figures cannot be equal to the total number of items possessed by all the 6 persons?

(a) 41 (b) 47
(c) 53 (d) 58

Ans. (d) : According to question–

A = C + 3
D = B – 4
E = F – 6
C = E + 2
F = D + 3

Also it can be inferred that A = B – 2

$\therefore$ total no. of items = A + B + C + D + E + F

= A + (A + 2) + (A – 3) + (A – 2) + (A – 5) + (A + 1)

= 6A – 7

If A = 8 then total no. of items = 8 × 6 – 7 = 41

For A = 9 then total no. of items = 9 × 6 – 7 = 47

For A = 10 then total no. of items = 10 × 6 – 7 = 53

$\therefore$ 58 can not be equal to the total number of items possessed by all the 6 persons.

142. Ten identical particles are moving randomly inside a closed box. What is the probability that at any given point of time all the ten particles will be lying in the same half of the box?

(a) 1/2 (b) 1/5
(c) $1/2^9$ (d) 2/11

Ans. (c) : Probability of a particle lying in any particular half = 1/2

$\therefore$ Probability of all 10 particles lying in either

$$\text{half} = \left(\frac{1}{2}\right)^{10} + \left(\frac{1}{2}\right)^{10} = 2\left[\frac{1}{2}\right]^{10} = \frac{1}{2^9}$$

143. An equilateral triangular plate is to be cut into n number of identical small equilateral triangular plates. Which one of the following can be possible value of n?

(a) 196 (b) 216
(c) 256 (d) 296

Ans. (c) : An equilateral triangle whenever divided into equal parts, gives 4 equilateral triangles. Hence it is always its power of 4 whenever it is equally divided further.

Thus only 4^4 = 256 satisfy the above condition.

144. There are 10 identical coins and each one of them has 'H' engraved on its one face and 'T' engraved on its other face. These 10 coins are lying on a table and each one of them has 'II' face as the upper face.